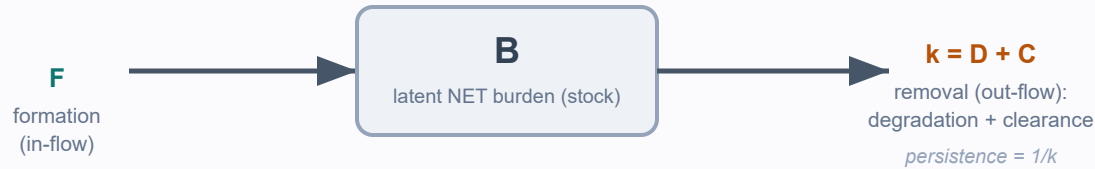


A two-layer observation model for NET burden

Why a single accumulated stock cannot identify the flows that generate it — two stacked sources of non-identifiability

1 · Process layer — the stock and its flows



$$dB/dt = F - kB$$

$$B^* = F/k$$

one scalar B identifies F/k ,
not F and k separately

excess formation (high F) and impaired removal (low k) are **observationally equivalent** at a single time point
steady state is the most favourable case; the realistic non-stationary case (acute COVID-19 / sepsis transients) is strictly worse

B is never observed directly

2 · Observation layer — the measured signal



$$Y = K_R \cdot B + \epsilon$$

K_R = detectability,
 ϵ = measurement noise

a **second, stacked** source of non-identifiability: even a known B need not be faithfully reflected in Y

Measurement breakers — what to co-measure to make a mechanism identifiable (design requirements, not demonstrated mechanisms)

formation assay +
serum degradation assay

time-course /
decay design

PTM-rate / enzyme
activity + total material

paired functional
NET-formation readout

paired metabolic +
functional (perturbation)

Adding latent processes without adding measurements cannot resolve the ambiguity — only preserve or worsen it; the remedy is to measure more, not to model more.

Figure 2 · an observation model, not a clinical or patient-calibrated model · synthetic illustrative trajectories are supplementary only

